

JTMK Project Report:

Smart Robotic Arm Control System

Project Title:

Smart Robotic Arm Control System for Object Sorting

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Introduction:

The Smart Robotic Arm Control System is a project designed to automate the process of sorting objects based on size and color using sensors and programmed robotic movement. The main goal is to simulate a real-world industrial automation process using cost-effective components suitable for learning environments.

Objectives:

1. To build a robotic arm using Arduino microcontroller and servo motors.
2. To detect and sort objects using IR and color sensors.
3. To automate movement sequences using precise logic control.
4. To enhance students' understanding of robotics and automation integration.

Tools and Technologies Used:

- Hardware: Arduino Uno, Servo Motors (x4), Color Sensor (TCS3200), IR Sensors, Breadboard, Jumper Wires.
- Software: Arduino IDE, Fritzing (for circuit design), TinkerCAD (simulation).
- Programming Language: C/C++ for Arduino.

Features:

- Object detection and classification using real-time sensors.
- Robotic arm performs pick-and-place tasks based on logic flow.
- LCD screen to display sorting status.
- Emergency stop button integrated for safety simulation.

Circuit Design Diagram:

(Insert image here of circuit diagram or simulated design)

Implementation Process:

1. Planning: Designed the structure and flowchart for the sorting process.
2. Hardware Assembly: Built the robotic arm and connected sensors.
3. Programming: Wrote and tested the Arduino code for accurate movement.
4. Testing: Ran multiple tests with objects of various sizes and colors.
5. Improvements: Calibrated sensor sensitivity and arm precision.

Results:

- Successfully sorted red, green, and blue objects.
- Sorting speed: 1 object per 5 seconds.
- Accuracy rate: 92% under good lighting conditions.

Challenges:

- Color sensor misread under poor lighting.
- Servo motor overheating after long runs.
- Balancing the arm structure to avoid tipping.

Future Enhancements:

- Add object weight detection using pressure sensors.
- Use a more advanced microcontroller (e.g., Raspberry Pi).
- Implement wireless control via Bluetooth or Wi-Fi.

Conclusion:

This project enhanced our understanding of robotics, programming, and sensor integration. The robotic arm effectively simulates real-life sorting mechanisms used in industrial automation and can be further developed for more complex tasks.

References:

1. Arduino Official Docs - <https://www.arduino.cc/en/Guide>
2. TCS3200 Color Sensor Guide - <https://lastminuteengineers.com/tcs230-tcs3200-color-sensor-arduino-tutorial/>
3. Servo Motor Control - <https://www.electronicwings.com/arduino/servo-motor-interfacing-with-arduino-uno>
4. IR Sensor Basics - <https://components101.com/sensors/ir-sensor-module>